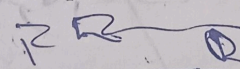
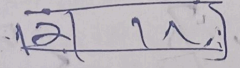
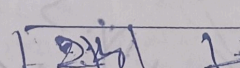
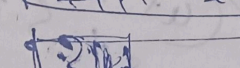
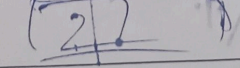


Instructions:	1	Missing data, if any, may be suitable assumed.
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Q.No	PART A (Compulsory question)	[L]	[CO]	[PO]	[M]
1.	a. Name the Program Outcomes (POs) ,which are related to the fundamental operations of a stack, such as push, pop, and display? b. What is Structure in C? Explain the different ways of declaring structures with examples. c. Write C program to find sum of elements of array using pointer.	2	1,2	1,2,3	2 5 3
PART B (Answer any 1 questions)					
2.	Identify the data structure required to convert the given infix expression to postfix expression and apply the same to convert the following expression. $(A + B) * (C - (D / E \wedge F) * G) + H$	3	3	2,3	10
3.	Develop a program in C that defines a structure for Employee containing employee ID, name, and salary. The program should allow the user to input details of N employees and then sort them by salary in ascending order.	3	1	2	10
PART C (Answer any,1 question)					
4.	You are working in a logistics company that manages shipments. Each shipment is processed in a First In, First Out (FIFO) manner. Your task is to examine how the process works in different operations and track the status after each action. The maximum capacity of the queue is 3: i. display()  ii. enqueue(2)  iii. enqueue(4)  iv. dequeue()  v. rear()  vi. enqueue(8)  vii. peek() 	4	3	2,3	10

viii. enqueue(7)

ix. dequeue()

x. enqueue(11)

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5.

You are working as a software developer for a company that builds a game AI. The AI needs to evaluate mathematical expressions provided in postfix notation to make real-time decisions during gameplay. Since postfix expressions don't need parentheses to handle operator precedence, it's ideal for the fast-paced environment in which the AI operates.

You've been given the following postfix expression that represents a calculation needed for the AI to decide the next move. Your task is to analyze this expression step-by-step, using a suitable data structure, and arrive at the result so that the next move can be decided.

6 2 3 + - 3 8 2 / + *

4

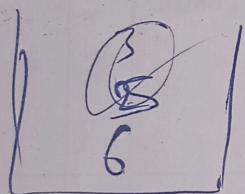
3

2,3

10

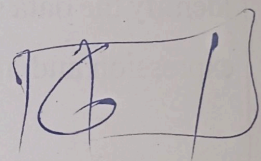
$$\text{rear} = (\text{rear} + 1) \% 5$$

3 4 5 1 2



$$2 + 3 = 5$$
$$6 - 5 = 1$$

$$\text{rear} = (\text{rear} + 1) \% 5$$

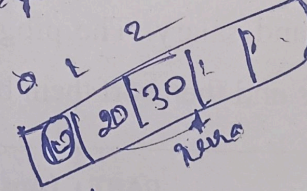


Sum

Not Exp

6

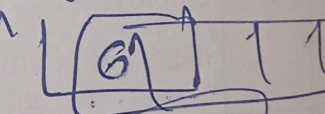
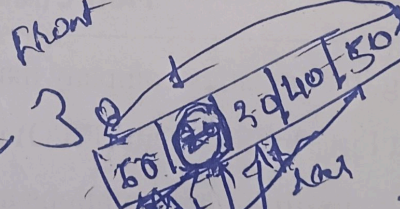
6



Stack

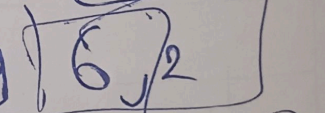
2

6, 2



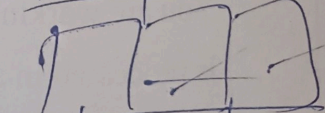
3

6, 2, 3



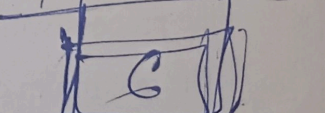
+

6, 2 + 3



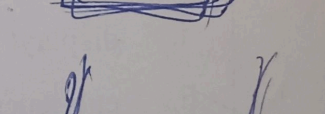
-

6, 5, 3



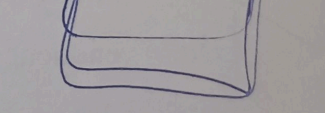
8

6, 5, 3



2

6, 5, 3



2

6, 5, 3

